MIZORAM PUBLIC SERVICE COMMISSION

COMPETITIVE EXAMINATION (COMBINED TECHNICAL) FOR RECRUITMENT TO THE POSTS OF AGRICULTURE & ALLIED SERVICES, GOVERNMENT OF MIZORAM.

(NOVEMBER-2025)

AGRICULTURE SCIENCE PAPER-II

Time Allowed : 3 hours FM : 200

SECTION - A *(Multiple Choice questions) (100 Marks)*

All questions carry equal mark of 2 each. Attempt all questions.

*This Section should be answered only on the **OMR Response Sheet** provided.*

Notes regarding Negative Marking :

(i) There will be penalty (negative marking) for every wrong answer given by a candidate in the MCQ type questions.

(ii) For each question for which a wrong answer has been given by the candidate, one-third of the marks assigned to that question will be deducted as penalty

(iii) If a candidate gives more than one answer to one question, it will be treated as a wrong answer even if any one of the given answers happens to be correct and there will be same penalty as above for that question.

(iv) If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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1. Who is the father of Plant Pathology?

(a) TJ Burill (b) Anton de Bary

(c) Needham (d) EJ Butler

2. 'White blisters of crucifers' is caused by pathogen.

(a) *Pythium debaryanam* (b) *Sclerospora sorghi*

(c) *Albugo candida* (d) *Plasmopora viticola*

3. The disease damping off seedlings during nursery stage is caused by the pathogen

(a) *Physoderma maydis* (b) *Fusarium spp*

(c) *Pythium spp* (d) *Colletotrichum lindemuthianum*

4. Damping off and leaf blights are effectively checked by:

(a) Bordeaux mixture (b) Thiram

(c) Burgundy mixture (d) Copper oxychloride

5. The specific fungicide for management of Blast of Rice is:

(a) Vitavax (b) Carbendazim

(c) Tricyclazole (d) Benlate

6. Bunchy top of Papaya (caused by MLO) is spread by:

(a) Nematodes (b) Whitefly

(c) Fungi (d) Leaf hoppers

7. Bacterial diseases are controlled by use of chemicals.

(a) Kelthane (b) Antibiotics

(c) Fungicide (d) Viricides

8. The Fungi which transmit Plant Viruses belong to class.

(a) Basidiomycetes (b) Zygomycetes

(c) Oomycetes (d) Plasmodiophoromycetes

9. Early blight of Potato produces:

(a) Conidia (b) Uredia

(c) Telia (d) Acecia

10. Most widely used fungicide for smut fungi is:

(a) Vitavax (b) Dithane M-45

(c) Plantvax (d) Dithane Z-78

11. Which of the following fungicide is not systemic in nature?

(a) Vitavax (b) Benlate

(c) Thiram (d) Topsin

12. Soil micro-organisms are most active at:

(a) 18 - 20°C (b) 30 - 32°C

(c) 26 - 28°C (d) 34 - 36°C

13. What is the source of Agar Agar?

(a) Bacteria (b) Mycoplasma

(c) Fungi (d) Algae

14. Which is the most recent widely used fungicide for the control of Powdery mildew?

(a) Calaxin (b) Sulphur dust

(c) Dithane M-45 (d) Apron SD-35

15. The major storage fungi that affects the food grain is:

(a) Rhizobium (b) Cercospora

(c) Mucor (d) Aspergillus

16. Systematic hybridization on a scientific basis began with the work.

(a) Shull – Maize (b) Patel – Cotton

(c) koelreuter -Nicotiana (d) Knight – Fruit

17. Emasculation of Flower is necessary for hybridization in:

(a) Male sterile plants (b) Self pollinated plants

(c) Self incompatible plants (d) Dioecious plants

18. In Green plants, photosynthesis takes place in:

(a) Proplastids (b) Leucoplasts

(c) chromoplasts (d) Chloroplasts

19. The term heterosis was introduced by:

(a) Davenport (1908) (b) Shull (1914)

(c) Hull (1945) (d) East (1936)

20. Progeny of a single plant obtained by asexual reproduction is:

(a) Pure line (b) Inbred

(c) Clone (d) Strain

21. In maize, cross pollination occurs due to:

(a) Protandry (b) Self - incompatibility

(c) Protogyny (d) Genetic diversity

22. Bulk population breeding is suitable for:

(a) Fruit crops (b) Vegetable crops

(c) Small grain crops (d) Flower crops

23. The process of pollen grains transferred from the anthers of a flower to the stigma of the same flower is called:

(a) In vitro fertilization (b) Autogamy

(c) Xenogamy (d) Geitonogamy

24. The crossing of F1 to either of the parents is known as:

(a) Test cross (b) F1 cross

(c) Back cross (d) Interspecific crossing

25. Genetically most pure seed is:

(a) Breeder seed (b) Foundation seeds

(c) Nucleus seed (d) Certified seed

26. Multiple effects of a single gene is known as:

(a) Pleiotropy (b) Heteroploidy

(c) Polyploidy (d) Monoploidy

27. The oldest method of plant breeding is:

(a) Introduction (b) Hybridization

(c) Selection (d) Back crossing

28. The term self - incompatibility was coined by:

(a) Harrington (b) Stout

(c) Lewis (d) East

29. The concept of male sterility was given by:

(a) Jones and davis (b) Nilsson and Ehle

(c) Shull (d) Jenson

30. Which of the following method of plant breeding is not used for asexually propagated crops?

(a) Plant introduction
(b) Mutation breeding
(c) Clonal selection
(d) Pure line selection

31. Pests outbreaks tend to occur when:

(a) Crops are planted in monoculture
(b) Natural enemies are imported from abroad
(c) Farmers switch to new crops
(d) Mixed cropping

32. Biological control is likely to be most effective when the predator or parasite has a:

(a) Long life cycle
(b) High reproductive rate
(c) Wide range of preferred hosts
(d) Short life cycle

33. Which control strategy is likely to have the greatest impact on non-target organisms?

(a) Chemical control
(b) Cultural control
(c) Biological control
(d) Physical control

34. Which class of chemical insecticides is characterized by a relatively high degree of environmental persistence?

(a) Carbamates
(b) Synthetic pyrethroids
(c) Organophosphates
(d) Chlorinated hydrocarbons

35. Which of the following is systemic insecticide?

(a) Methamidophos
(b) Cypermethrin
(c) Imidachloprid
(d) Deltamethrin

36. The most important physical factor which affects the life of an insect is:

(a) Temperature
(b) Light
(c) Humidity
(d) Rainfall

37. Pheromones used against the fruit fly is:

(a) Methyle eugenol
(b) Carniform
(c) Methyl bromide
(d) Carbofuran

38. Following insecticide is safely recommended on vegetables:

(a) Carbaryl
(b) Malathion
(c) Aldicarb
(d) Endrin

39. The process of leaving off the colony by the queen is known as:

(a) Absconding
(b) Supersedure
(c) Swarming
(d) Queen excluder

40. Which pesticide is suitable for controlling the white grub?

(a) Malathion
(b) Phorate
(c) Chlorpyriphos
(d) Cypermethrin

41. Stomata open and close due to:

(a) Circadian rhythm
(b) Pressure of gases inside the leaves
(c) Genetic clock
(d) Turgor pressure of guard cells

42. The core metal of chlorophyll is:

(a) Ni
(b) Fe
(c) Cu
(d) Mg

43. Which one of the following element is almost non-essential for plants?

(a) Zn
(b) Ca
(c) Na
(d) Mo

44. Which part of root absorb water and minerals?

(a) Root cap
(b) Epidermis
(c) Root hairs
(d) Endodermis

45. Apical dominance is the result of:

(a) Auxins
(b) Ethylene
(c) Cytokinins
(d) GA

46. The ideal time for air layering is:

(a) February - March
(b) May – June
(c) November- December
(d) July – August

47. High production in protected cultivation is due to:

(a) High photosynthetic efficiency
(b) High Oxygen
(c) High temperature
(d) Less CO_2 concentration

48. Indian Institute of Horticultural Research (IIHR) is located at:

(a) N. Delhi
(b) Hyderabad
(c) Bengaluru (Bangalore)
(d) Kolkata

49. Which crop is most susceptible to viral infection?

(a) Chillies
(b) Onion
(c) Tomato
(d) brinjal

50. Botanically pineapple fruit is:

(a) Sorosis
(b) Berry
(c) Balusta
(d) pome

SECTION - B *(Conventional Type) (100 Marks)*

*This Section should be answered only on the **Answer Sheet** provided.*
Marks for each question is indicated against it.
*Attempt **any five (5)** questions.*

1. What are the differences between Prokaryotic and Eukaryotic organisms? Explain Prokaryotic Bacterial Cell structure and functions with diagram indicating different types of structure. **(20)**

2. What are the principles of plant disease management? Elaborate different components of Integrated Disease Management (IDM) in plants with one example from each component. **(20)**

3. Answer any four (4) of the following in brief: **(4×5=20)**

(a) What is plant breeding and why it is important for crop improvements?

(b) Differences between cross-pollinated and self-pollinated crops.

(c) Differences between Hybridization and Selection procedures in plant breeding.

(d) Concept and importance of heterosis in crop production.

(e) Importance of honey bee in crop production.

(f) Difference between C_3 & C_4 cycle.

4. What are pests? Mention five (5) common insect pests of rice and explain the nature of damage, management strategies adopted for yellow stem borer infestation in rice. **(20)**

5. What are plant parasitic Nematodes? Describe the symptoms of Root-knot Nematodes in the infected plants and management practices for controlling their population in soil. **(20)**

6. Define Kitchen Gardening and describe the concept and importance of Kitchen Gardening for upliftment of rural economy. **(20)**

7. Explain the diagnostic symptoms of *Fusarium wilt* of Tomato, *Late blight* of Potato, *Rhizome rot* of Ginger and what are the practices adopted for effective management of soil borne disease incidence. **(20)**

8. Define Phytohormones and explain different types of phytohormones with their functions and its Agricultural applications. **(20)**

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